

Shanzay Jamil

, Pakistan | | [Linkedln](#) | [Github](#)

SUMAARY

I'm a Computer Engineering graduate with a strong interest in AI-driven cybersecurity, NLP, and IoT security. Through my projects and internships, I've worked on deploying ML models on embedded hardware, implementing encryption techniques in Python, and building NLP-based classification systems. These experiences pushed me toward wanting to do deeper research in intrusion detection, IoT security, and threat analysis using language models.

SKILLS

Technical Skills:

Python, Scikit-learn, TensorFlow, Pandas, Matplotlib, Power BI, DAX, (TCP/IP, CCNA), Kali Linux, Applied Cryptography SQL, Advanced in SQL, MongoDB, JavaScript , React.js, Node.js, PHP, HTML/CSS, Firebase, Styled Components,

EDUCATION

University of Engineering and Technology, Taxila Pakistan

Bachelor of Computer Engineering, **GPA: 3.59/4.0 (89.8%)**

WORK EXPERIENCE

Lab Instructor, Air University

(Sep 2024 – Present)

- Conduct labs for Electrical and Computer Engineering courses, supporting hands-on learning and practical experimentation.
- Guide students through programming, AI, embedded systems, and hardware-related lab activities.
- Supervise student projects, mentor teams, and assist in project planning and technical execution.
- Support assessment, reporting, and lab documentation to ensure smooth academic operations

Power BI Analyst ,NCRI Pakistan

(June 2024 – Sep 2024)

- Used Python (Pandas, NumPy) to clean, transform, and merge multi-source datasets as part of ETL workflows feeding business dashboards.
- Performed exploratory data analysis in Python to identify trends and support reporting decisions.

Research Engineer, Pi Invent, Islamabad

(Aug 2023 – Oct 2023)

- Researched modern cryptographic algorithms and security protocols, with focus on symmetric and asymmetric encryption schemes.
- Implemented encryption/decryption pipelines in Python, including building a working prototype for one algorithm from scratch to study its performance trade-offs.
- Benchmarked encryption/decryption speed and security strength across approaches and documented findings for the team.

Data Science Intern ,UET Taxila, Pakistan

(Aug 2022 – Feb 2023)

- Collected and prepared speech-command datasets using Edge Impulse.
- Trained a keyword-spotting model ($\approx 89\%$ accuracy) for embedded devices.
- Deployed ML models on edge hardware using TinyML frameworks.
- Documented results and supported model tuning and dataset improvement.

ACADEMIC PROJECTS

Keyword Spotting & Voice-Based Laptop Control using TinyML

- Collected and curated a custom voice-command dataset from university volunteers for a speech-recognition pipeline.
- Used Edge Impulse for audio tokenization, feature extraction (MFCC), data clustering, and model training.
- Trained a lightweight keyword-spotting model achieving $\sim 89\%$ accuracy, optimized for low-power inference.
- Embedded and deployed the trained model onto an Arduino Uno using TinyML, enabling real-time, on-device voice control of laptop functions without cloud dependency.

Network Intrusion Detection System using ML (CIC-IDS2017, Python, Scikit-learn)

- Built a binary classification pipeline to detect malicious network traffic using the **CIC-IDS2017** benchmark dataset
- Trained and compared Random Forest and Neural Network classifiers across 7 attack categories including DDoS, DoS, PortScan, and Web Attacks.

- Performed feature importance analysis identifying Flow Duration, Packet Length, and Inter-Arrival Time as the most discriminative behavioral signals for threat detection.

Language Identifier using NLP

- Built a text classification system to automatically detect the language of input text using tokenization and feature extraction.
- Trained and compared multiple ML classifiers (e.g. Naive Bayes, SVM) to identify the best-performing model.
- Evaluated model accuracy and generalization across multiple multilingual datasets, refining preprocessing to improve robustness on short text inputs.

Posture Correction System using MediaPipe

- Built a real-time computer-vision system that detects human body landmarks using MediaPipe to identify incorrect sitting and standing postures.
- Designed feedback logic to assist users in improving posture and reducing strain.

Body Measurements App using Android Studio & Machine Learning (FYP)

- Developed a mobile app that estimates human body measurements using computer-vision techniques and ML-based prediction models.
- Implemented feature extraction, preprocessing, and model tuning to improve accuracy and user experience.

T20 World Cup Data Analysis (Python, Pandas, Power BI)

- Performed exploratory data analysis on cricket performance data to uncover match trends and player insights. Built interactive dashboards in Power BI to visualize metrics and performance indicators.

CERTIFICATIONS

- **Introduction to TensorFlow (DeepLearning.AI)**
Building and deploying AI, Machine Learning, and Deep Learning models using TensorFlow.
- **Foundation of Data Science (Google)**
Data planning, workflow best practices, and sourcing data across multiple resources.
- **Get Started with Python (Google)**
Data cleaning, visualization, and analysis using Python and Seaborn.
- **IBM Professional Data Analysis (Courses 1–3)**
Introduction to Data Analytics, Excel for Data Analysis, Data Visualization with Excel and Cognos.